10/19/2026: Multiple Proofs

CSCI 246: Discrete Structures

Textbook reference: Ch. 2, Hampkins

Today's Agenda

- Proof Jigsaw
- (If time) Review previous group exercises

Why multiple proofs?

Once you've proved a theorem, you know it's true. So why would you ever want a second proof?

Why multiple proofs?

Once you've proved a theorem, you know it's true. So why would you ever want a second proof?

1. **Confidence.** Two independent arguments make an error less likely (like testing code against a second implementation).

Why multiple proofs?

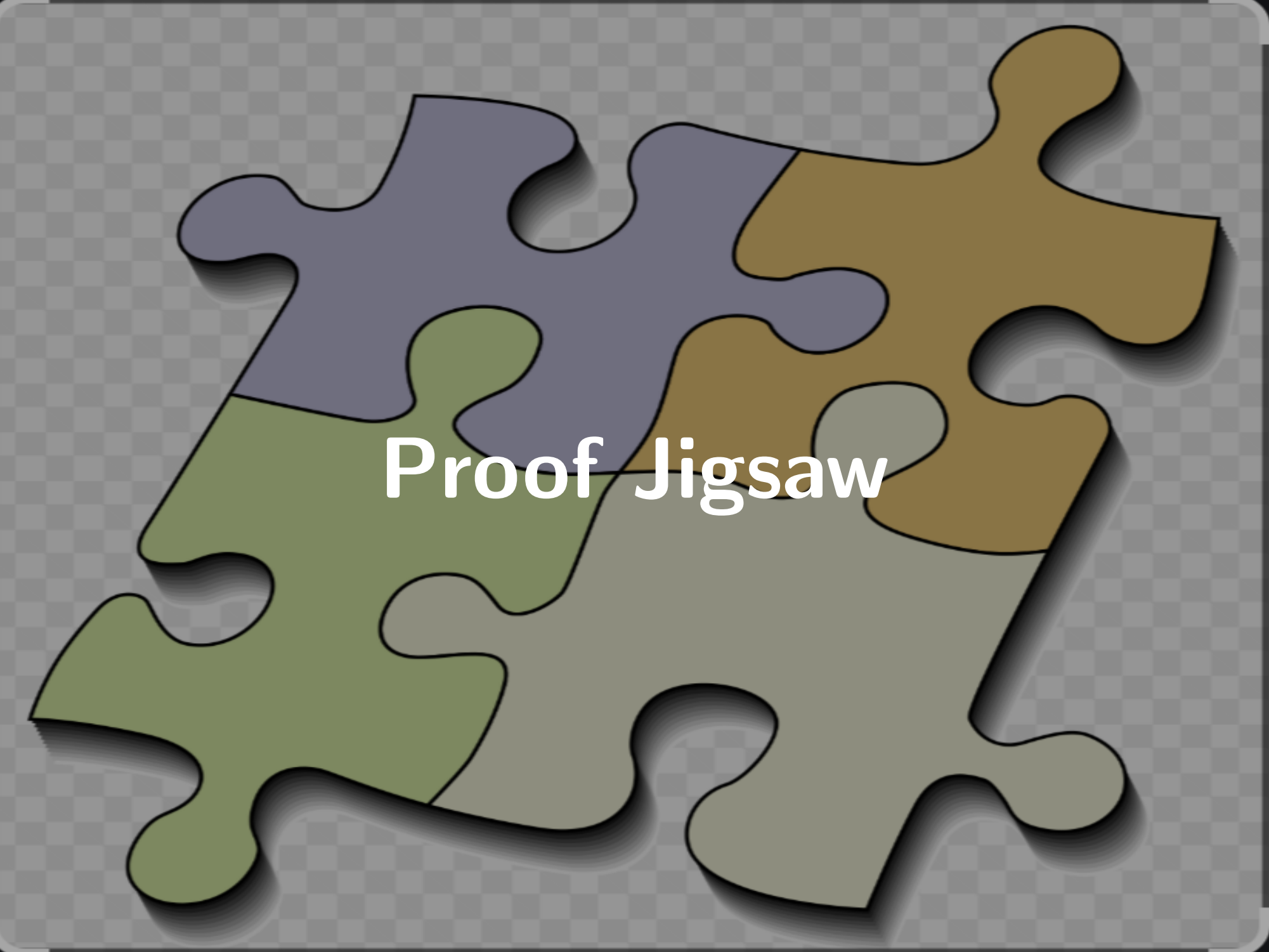
Once you've proved a theorem, you know it's true. So why would you ever want a second proof?

1. **Confidence.** Two independent arguments make an error less likely (like testing code against a second implementation).
2. **Understanding.** Different proofs explain *why* the theorem is true from different angles (cases, algebra, pictures, counting, ...).

Why multiple proofs?

Once you've proved a theorem, you know it's true. So why would you ever want a second proof?

1. **Confidence.** Two independent arguments make an error less likely (like testing code against a second implementation).
2. **Understanding.** Different proofs explain *why* the theorem is true from different angles (cases, algebra, pictures, counting, . . .).
3. **Generalization.** Different proofs suggest different generalizations. (Hamkins' main point.)


Proof Jigsaw

Theorem

For any natural number n, the number $n^2 - n$ is even.

- **Proof #1: By Cases.** Miller, Maverick; Moss, Emily; Neuman, Evan; Thompson, Jaiden; Vath, Ethan
- **Proof #2: By High-School Algebra.** Lee, Holden; Fellows, Audrey; Horner, Jayden; Mousseau, Matthew; Roberts, Matthew
- **Proof #3: By Induction.** Blaisdell, Olyver; Fuller, Ryan; Hager, Nathan; Kintzel, Samantha; Mcbride, Daniel
- **Proof #4: By Geometrical Figure.** Ashley, Isabelle; Bailey, Tanna; Cardenas, Dominic; Fabik, Roland; Shevtsov, Timothy
- **Proof #5: By Combinatorics.** Griffen, Clara; Grutsch, Tanner; Mayala, Reece; Ramos, Vydel; Roller, Nathan
- **Proof #6: By Modular Arithmetic.** Cutler, George; Fitzgerald, Ryan; McLean, Connor; Putnam, John; Thiessen, Logan
- **Proof #7: By Summation Identity.** Johnson, Jaret; Lameres, Alexis; Rathbun, Jedidiah; Ryan, Otis; Stensland, Emma

Extensibility

Which of your proofs still works if we replace 2 with 3? With 4?